

Risk Assessment and Management

Introduction

Risk assessment and management is an important process in software development projects. It helps identify potential risks that may negatively affect the development, performance, and security of the system. In the StockWise inventory management system, risk management is used to detect possible problems early and reduce their impact on the project.

Identified Risks in the StockWise Project

Several potential risks have been identified during the planning and development of the StockWise system:

1. Data Loss Risk

Inventory data stored in the database may be lost due to database failures, incorrect operations, or system crashes.

2. Security Risk

Unauthorized users may attempt to access the system or sensitive information such as inventory records and user credentials.

3. System Performance Risk

The system may experience slow response times or performance issues when handling large amounts of inventory data or multiple user requests.

4. Integration Risk

Problems may occur during the integration of the frontend (React) and backend (Spring Boot) components of the system.

5. User Error Risk

Users may accidentally delete or modify product or inventory information due to incorrect system usage.

6. Server or Deployment Risk

System downtime or hosting problems may affect system availability.

Risk Evaluation

Each risk is evaluated based on two factors:

- **Probability:** The likelihood of the risk occurring
- **Impact:** The potential damage the risk may cause to the project or system

This evaluation helps the project team prioritize risks and focus on the most critical ones.

Risk Management Approach

To manage the identified risks, the StockWise development team applies several strategies during the project lifecycle. These strategies include regular system monitoring, secure coding practices, database backup mechanisms, and proper authentication and authorization controls.

By applying these risk management practices, the project aims to ensure system reliability, protect sensitive data, and maintain stable inventory tracking operations.